Spring Framework 6

Beginner to Guru

Introduction to Spring Application Events

Introduction to Spring Application Events

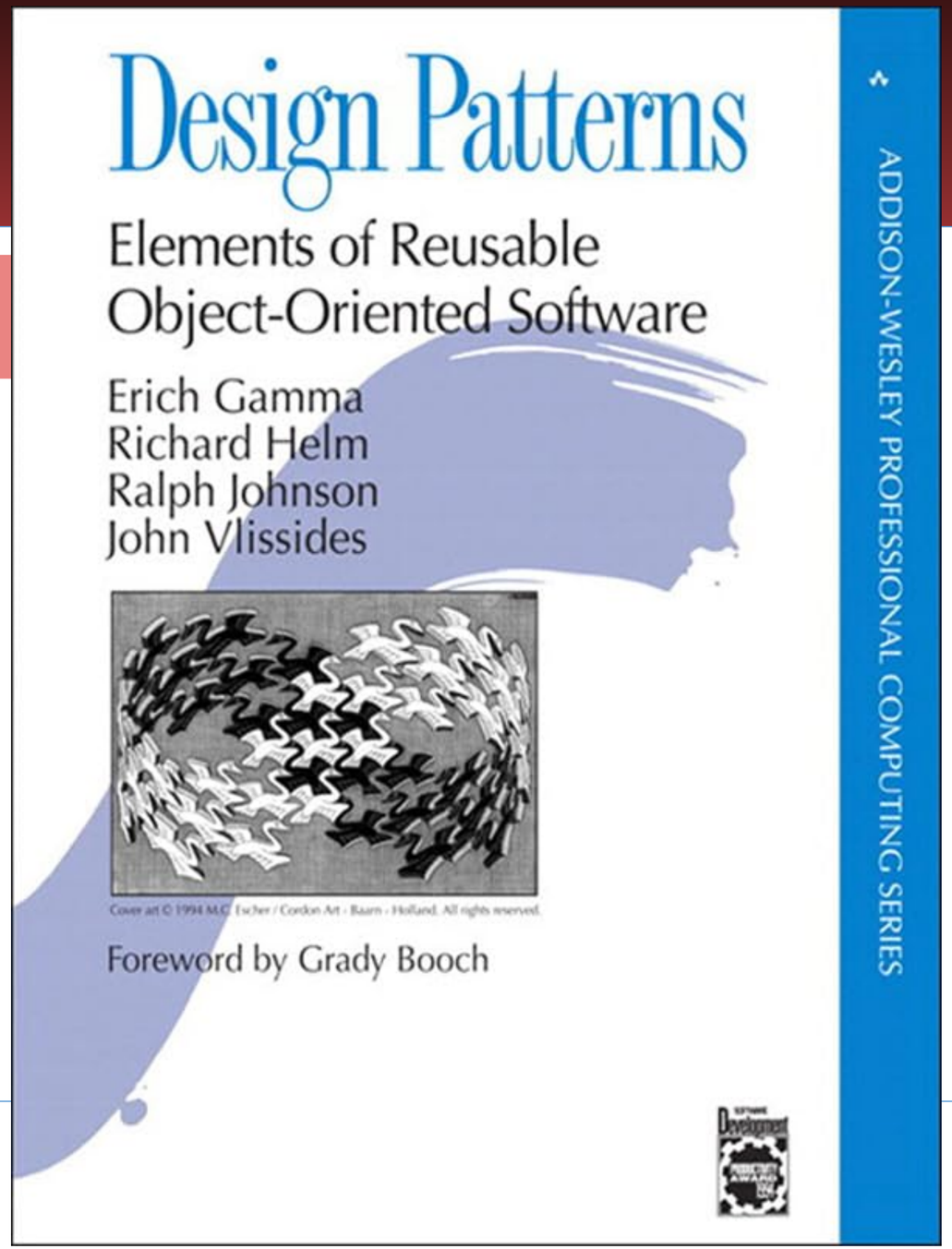
- Spring Events:
  - **System Events** - Subscribable Events in the Application Context
    - Typically Life Cycle Type Events, or OS Level Events
  - **Application Events** - Subscribable Events Defined in the Application
- Application Events Allow you to Publish and Event for Zero, One or more Subscribers
- By Default are Synchronous
- Optionally can be Asynchronous

Introduction to Spring Application Events

- Example Use Cases:
  - Sending an Email
  - Recording to an Audit Log
  - Notifying One or Multiple Consumers of a State Change
  - Recording Login Attempts
  - Post Processing - such as Fraud Detection

Observer Pattern

- Spring Events Implement the Observer Pattern
- Originally Defined in Design Patterns
  - Authors Nick-Named Gang of Four (GoF)
- Classic Engineering Book
  - While on C++, still relevant for OOP with Java
- Spring Framework Guru Blog has Java examples of all Design Patterns for Further Learning



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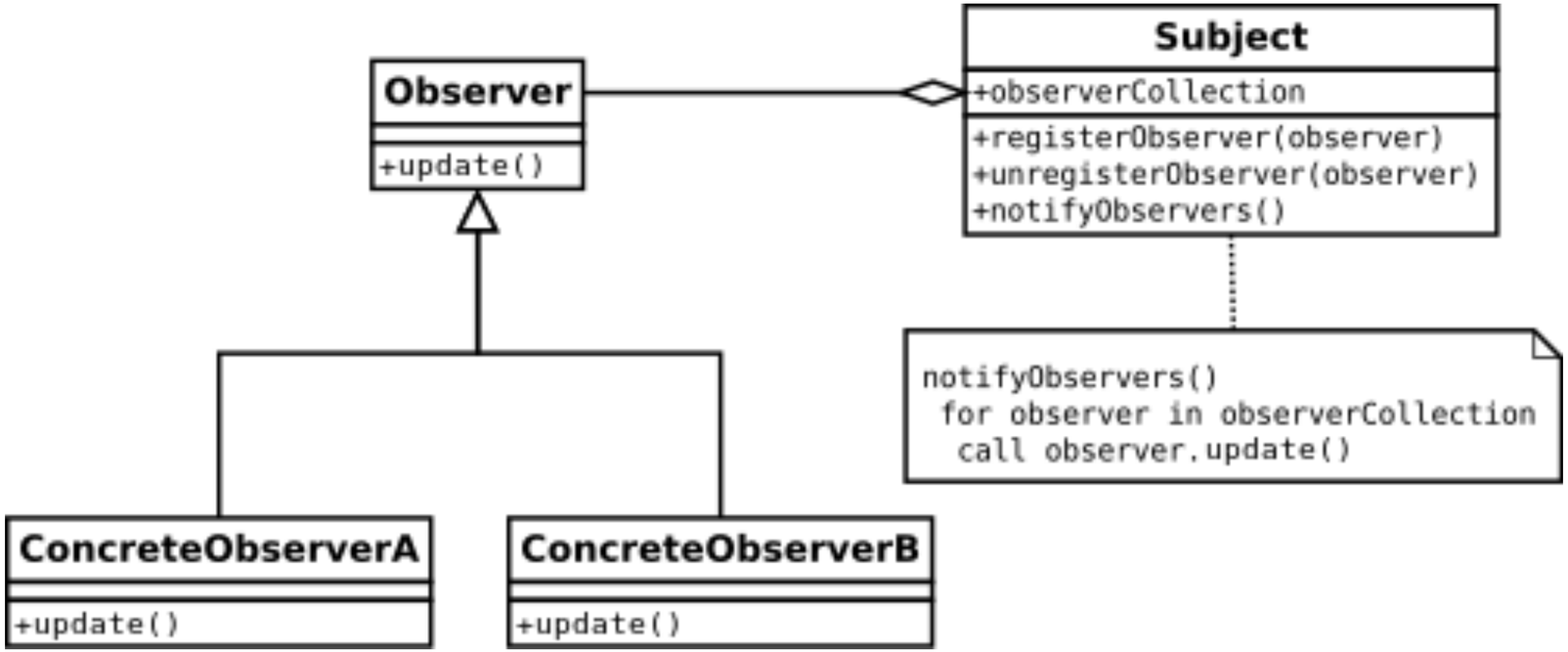
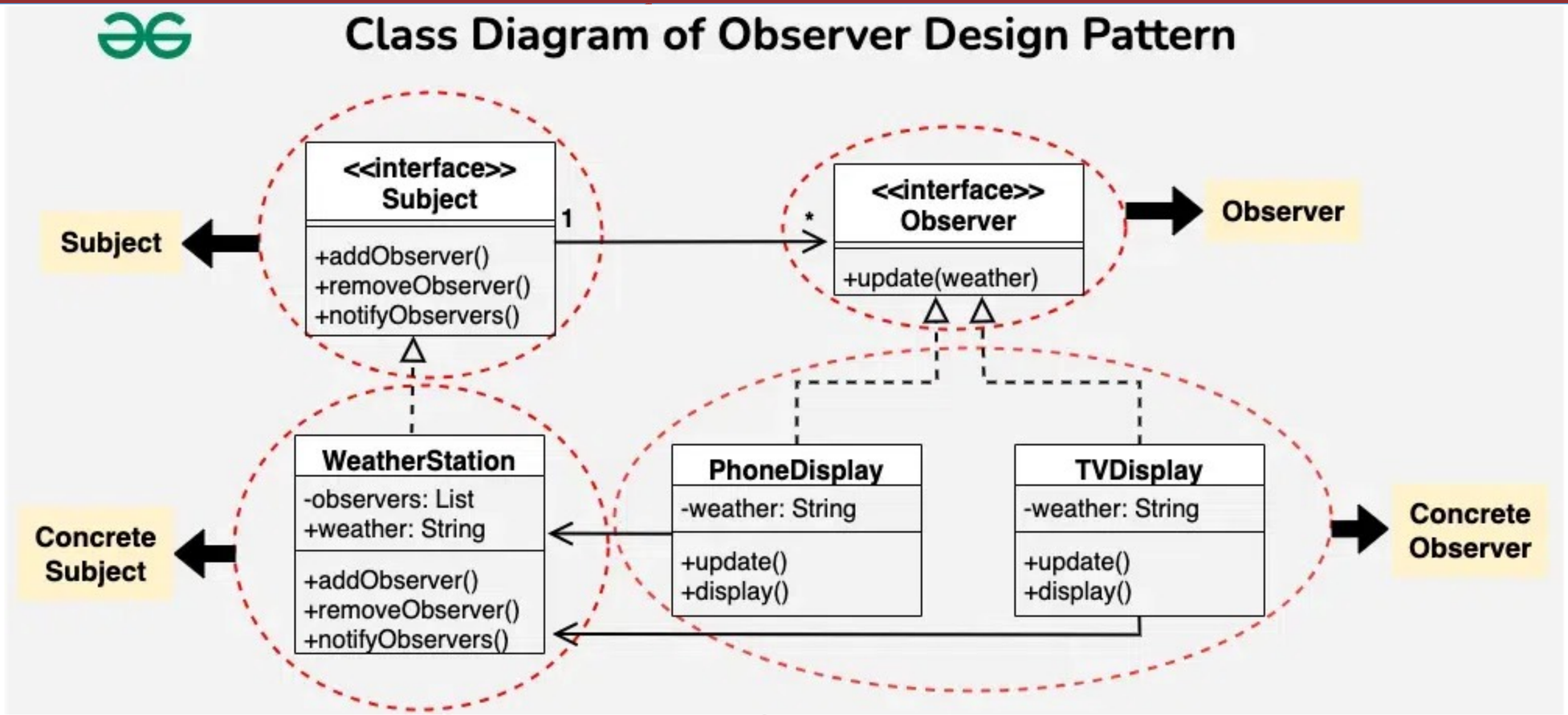
Observer Pattern





Source: GeeksforGeeks

Application Events in Spring

- Events are Defined as Java POJOs
- Events are Published via Springs Application Event Publisher
- Events are Subscribed to by:
  - Components Implementing the AplicationEventPublisherAware Interface
  - Components Implementing the ApplicationListener Interface
  - Components with Methods Annotated with @EventListener

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